



TEST REPORT GEISA CONFORMANCE TESTS

Receiver: GEISA

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Tests summary

- Total number of tests: 105
- Total number of passed tests: 100
- Total number of failures: 0
- Total number of skipped: 5

Test reports

Tests GEISA LEE Conformance tests installation

Tests	Results
Checking conformance tests installation	PASS
Checking conformance tests configuration file	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA LEE Operating System

Test ID	Tests	Results
GEISA-LEE-OS-SHALL-01	GEISA LEE SHALL use Linux as the core operating system.	PASS
GEISA-LEE-OS-SHOULD-02	The LEE platform implementer SHOULD use a Linux 6.x or greater kernel.	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA LEE Application Isolation

Test ID	Tests	Results
GEISA-LEE-APP-MUST-01	The platform MUST provide a base image	PASS
GEISA-LEE-APP-MUST-02	The platform MUST accept application images	PASS
GEISA-LEE-APP-SHALL-03	GEISA applications SHALL be isolated from each other as described in /system-architecture.	PASS
GEISA-LEE-APP-SHALL-04	The platform SHALL control the number of CPUs allocated to a GEISA application.	PASS
GEISA-LEE-APP-SHALL-05	The platform SHALL control the percentage of CPU allowed to a GEISA application.	PASS
GEISA-LEE-APP-SHALL-06	The platform SHALL control the priority of a GEISA application.	PASS
GEISA-LEE-APP-SHALL-07	The platform SHALL control the maximum RAM allocated to a GEISA application.	PASS
GEISA-LEE-APP-SHALL-08	The platform SHALL control a GEISA application's persistent storage	PASS
GEISA-LEE-APP-SHALL-09	The platform SHALL control a GEISA application's non-persistent storage	PASS
GEISA-LEE-APP-SHALL-10	The platform SHALL provide standard socket-based IP connectivity when granted in the operator manifest	SKIPPED
GEISA-LEE-APP-SHALL-11	The platform SHALL enforce the network access control list described in the application manifest	SKIPPED
GEISA-LEE-APP-SHALL-12	The platform SHALL enforce the network volume limits described in the application manifest	SKIPPED

- number of tests: 12

- number of failures: 0
- number of skipped: 3

Tests GEISA LEE Linux Base Libraries

Test ID	Tests	Results
GEISA-LEE-LIB-MUST-01	Other C Libraries (MUST): libasyncns	PASS
GEISA-LEE-LIB-MUST-02	Other C Libraries (MUST): libatomic	PASS
GEISA-LEE-LIB-MUST-03	Other C Libraries (MUST): libcrypto	PASS
GEISA-LEE-LIB-MUST-04	Other C Libraries (MUST): libutil	PASS
GEISA-LEE-LIB-MUST-05	Other C Libraries (MUST): libz	PASS
GEISA-LEE-LIB-MUST-06	Other C Libraries (MUST): libmosquitto	PASS
GEISA-LEE-LIB-MUST-07	Other C Libraries (MUST): libprotobuf	PASS

- number of tests: 7
- number of failures: 0
- number of skipped: 0

Tests GEISA LEE Base Filesystem

Test ID	Tests	Results
GEISA-LEE-FS-MUST-01	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/bin).	PASS
GEISA-LEE-FS-MUST-02	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/dev).	PASS
GEISA-LEE-FS-MUST-03	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/etc).	PASS
GEISA-LEE-FS-MUST-04	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/home).	PASS
GEISA-LEE-FS-MUST-05	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/lib).	PASS
GEISA-LEE-FS-MUST-06	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/proc).	PASS
GEISA-LEE-FS-MUST-07	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/sys).	PASS
GEISA-LEE-FS-MUST-08	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/tmp).	PASS
GEISA-LEE-FS-MUST-09	A skeleton filesystem MUST be populated including typical paths for binaries and libraries (/var).	PASS
GEISA-LEE-FS-MUST-10	A skeleton filesystem MUST be populated with typical files including at a minimum (/etc/group).	PASS
GEISA-LEE-FS-MUST-11	A skeleton filesystem MUST be populated with typical files including at a minimum (/etc/hostname).	PASS
GEISA-LEE-FS-MUST-12	A skeleton filesystem MUST be populated with typical files including at a minimum (/etc/hosts).	PASS

Test ID	Tests	Results
GEISA-LEE-FS-MUST-13	A skeleton filesystem MUST be populated with typical files including at a minimum (/etc/passwd).	PASS
GEISA-LEE-FS-MUST-14	A skeleton filesystem MUST be populated with typical files including at a minimum (/etc/hosts including a localhost entry).	PASS
GEISA-LEE-FS-MUST-15	The GEISA platform implementation MUST provide the following virtual filesystem mounts within the container environment (/home/geisa).	PASS
GEISA-LEE-FS-MUST-16	The GEISA platform implementation MUST provide the following virtual filesystem mounts within the container environment (/).	PASS
GEISA-LEE-FS-MUST-17	Each GEISA Application MUST be provisioned in its own container environment separate from other Applications.	PASS
GEISA-LEE-FS-MUST-18	The GEISA platform implementation MUST provide the following virtual filesystem mount within the container environment (/proc).	PASS
GEISA-LEE-FS-MUST-19	The GEISA platform implementation MUST provide the following virtual filesystem mount within the container environment (/dev).	PASS
GEISA-LEE-FS-MUST-20	The GEISA platform implementation MUST provide the following virtual filesystem mount within the container environment (/sys).	PASS
GEISA-LEE-FS-MUST-21	The GEISA platform implementation MUST provide the following virtual filesystem mount within the container environment (/tmp).	PASS
GEISA-LEE-FS-MUST-22	The root filesystem MUST be mounted read-only, or the application MUST lack write permission outside the writable paths.	PASS
GEISA-LEE-FS-MUST-23	If the root filesystem is read-write, the platform MUST enforce file and directory permissions appropriately.	SKIPPED
GEISA-LEE-FS-MUST-24	If the root filesystem is read-write, the platform MUST limit its growable size to the manifest non-persistent storage limit.	SKIPPED
GEISA-LEE-FS-MUST-25	/dev MUST contain at a minimum (console).	PASS
GEISA-LEE-FS-MUST-26	/dev MUST contain at a minimum (full).	PASS
GEISA-LEE-FS-MUST-27	/dev MUST contain at a minimum (log).	PASS
GEISA-LEE-FS-MUST-28	/dev MUST contain at a minimum (null).	PASS
GEISA-LEE-FS-MUST-29	/dev MUST contain at a minimum (random).	PASS
GEISA-LEE-FS-MUST-30	/dev MUST contain at a minimum (stderr).	PASS
GEISA-LEE-FS-MUST-31	/dev MUST contain at a minimum (stdin).	PASS
GEISA-LEE-FS-MUST-32	/dev MUST contain at a minimum (stdout).	PASS
GEISA-LEE-FS-MUST-33	/dev MUST contain at a minimum (urandom).	PASS
GEISA-LEE-FS-MUST-34	/dev MUST contain at a minimum (zero).	PASS
GEISA-LEE-FS-MUST-35	/dev/log MUST be backed by the platform to filter and direct logs into the platform logging mechanism.	PASS
GEISA-LEE-FS-MUST-36	The following environment variable MUST be set when invoking Application processes (SHELL).	PASS
GEISA-LEE-FS-MUST-37	The following environment variable MUST be set when invoking Application processes (HOME).	PASS
GEISA-LEE-FS-MUST-38	The following environment variable MUST be set when invoking Application processes (USER).	PASS
GEISA-LEE-FS-MUST-39	The following environment variable MUST be set when invoking Application processes (PATH).	PASS
GEISA-LEE-FS-MUST-40	/tmp MUST be limited in size as described in the Application Deployment Manifest.	PASS

Test ID	Tests	Results
GEISA-LEE-FS-MUST-41	/home/geisa MUST be limited in size as described in the Application Deployment Manifest.	PASS
GEISA-LEE-FS-MUST-42	/home/geisa MUST be persistent across application restarts.	PASS
GEISA-LEE-FS-SHALL-44	Platform SHALL be responsible for metadata integrity (filesystem integrity mechanism).	PASS

- number of tests: 43
- number of failures: 0
- number of skipped: 2

Tests GEISA LEE Conformance tests installation

Tests	Results
Checking conformance tests installation	PASS
Checking conformance tests configuration file	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA LEE Base Filesystem

Test ID	Tests	Results
GEISA-LEE-FS-MUST-43	/home/geisa MUST be persistent across device reboots.	PASS

- number of tests: 1
- number of failures: 0
- number of skipped: 0

Tests GEISA API Conformance tests installation

Tests	Results
Checking conformance tests installation	PASS
Checking conformance tests configuration file	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA API Connection to MQTT broker

Test ID	Tests	Results
GEISA-API-CONN-01	Checking connection to MQTT broker	PASS

- number of tests: 1
- number of failures: 0
- number of skipped: 0

Tests GEISA API Test discovery messages

Test ID	Tests	Results
GEISA-API-DISC-01	Checking discovery status	PASS
GEISA-API-DISC-02	Checking discovery geisa messages	PASS
GEISA-API-DISC-03	Checking discovery device messages	PASS
GEISA-API-DISC-04	Checking discovery operator messages	PASS
GEISA-API-DISC-05	Checking discovery metrology messages	PASS
GEISA-API-DISC-06	Checking discovery sensor messages	PASS
GEISA-API-DISC-07	Checking discovery network messages	PASS
GEISA-API-DISC-08	Checking discovery waveform messages	PASS

- number of tests: 8
- number of failures: 0
- number of skipped: 0

Tests GEISA API Test instantaneous messages

Test ID	Tests	Results
GEISA-API-INST-01	Checking instantaneous messages	PASS

- number of tests: 1
- number of failures: 0
- number of skipped: 0

Tests GEISA API Test waveform messages

Test ID	Tests	Results
GEISA-API-WAVE-01	Checking waveform subscription/unsubscription	PASS
GEISA-API-WAVE-02	Checking waveform already subscribed	PASS
GEISA-API-WAVE-03	Checking waveform not subscribed	PASS
GEISA-API-WAVE-04	Checking waveform invalid stream_id	PASS

Test ID	Tests	Results
GEISA-API-WAVE-05	Checking waveform unavailable stream_id	PASS

- number of tests: 5
- number of failures: 0
- number of skipped: 0

Tests GEISA API Test sensor messages

Test ID	Tests	Results
GEISA-API-SENSOR-01	Checking request all readable sensors	PASS
GEISA-API-SENSOR-02	Checking request not present sensor	PASS
GEISA-API-SENSOR-03	Checking request non readable sensor	PASS

- number of tests: 3
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conformance tests installation

Tests	Results
Checking conformance tests installation	PASS
Checking conformance tests configuration file	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conditions before package update

Tests	Results
Checking if UPDATE_STATE is INITIAL(0) before package update	PASS
Checking if ACTIVATION_STATE is DISABLED(0) before package activation	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conditions after package update

Tests	Results
Checking if package can be pushed to client	PASS
Checking if UPDATE_STATE is changed to DELIVERED(3) after package update	PASS
Checking if ACTIVATION_STATE is DISABLED(0) before package activation	PASS

- number of tests: 3
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conditions after package installation

Tests	Results
Checking if package can be installed on client	PASS
Checking if UPDATE_STATE is changed to INSTALLED(4) after package installation	PASS
Checking if ACTIVATION_STATE is DISABLED(0) before package activation	PASS

- number of tests: 3
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conditions after package activation

Tests	Results
Checking if package can be activated on client	PASS
Checking if ACTIVATION_STATE is ACTIVATED(1) after package activation	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conditions after package deactivation

Tests	Results
Checking if package can be deactivated on client	PASS
Checking if ACTIVATION_STATE is DEACTIVATED(0) after package deactivation	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Tests GEISA ADM Conditions after package uninstallation

Tests	Results
Checking if package can be uninstalled on client	PASS
Checking if UPDATE_STATE is reset to INITIAL(0) after package uninstallation	PASS

- number of tests: 2
- number of failures: 0
- number of skipped: 0

Compliance Matrix

Matrix GEISA-LEE-matrix.csv

Requirement	Test id	Status
GEISA LEE Application Isolation Requirements	GEISA-LEE-APP-MUST-01	PASS
	GEISA-LEE-APP-MUST-02	PASS
	GEISA-LEE-APP-SHALL-03	PASS
	GEISA-LEE-APP-SHALL-04	PASS
	GEISA-LEE-APP-SHALL-05	PASS
	GEISA-LEE-APP-SHALL-06	PASS
	GEISA-LEE-APP-SHALL-07	PASS
	GEISA-LEE-APP-SHALL-08	PASS
	GEISA-LEE-APP-SHALL-09	PASS
	GEISA-LEE-APP-SHALL-10	SKIPPED
	GEISA-LEE-APP-SHALL-11	SKIPPED
	GEISA-LEE-APP-SHALL-12	SKIPPED

Requirement	Test id	Status
GEISA LEE Base Filesystem Requirements		
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	GEISA-LEE-FS-MUST-38	PASS
	GEISA-LEE-FS-MUST-39	PASS
Requirement	Test id	Status
	GEISA-LEE-FS-MUST-40	PASS
	GEISA-LEE-FS-MUST-41	PASS
	GEISA-LEE-FS-MUST-42	PASS
	GEISA-LEE-FS-MUST-43	PASS
	GEISA-LEE-FS-SHALL-44	PASS
GEISA LEE Base Libraries Requirements	GEISA-LEE-LIB-MUST-01	PASS
	GEISA-LEE-LIB-MUST-02	PASS
	GEISA-LEE-LIB-MUST-03	PASS
	GEISA-LEE-LIB-MUST-04	PASS
	GEISA-LEE-LIB-MUST-05	PASS
	GEISA-LEE-LIB-MUST-06	PASS
	GEISA-LEE-LIB-MUST-07	PASS
GEISA LEE OS Requirements	GEISA-LEE-OS-SHALL-01	PASS
	GEISA-LEE-OS-SHOULD-02	PASS

Matrix GEISA-API-matrix.csv

Requirement	Test id	Status
GEISA API Connection Requirements	GEISA-API-CONN-01	PASS
GEISA API Discovery Requirements	GEISA-API-DISC-01	PASS
	GEISA-API-DISC-02	PASS
	GEISA-API-DISC-03	PASS
	GEISA-API-DISC-04	PASS
	GEISA-API-DISC-05	PASS
	GEISA-API-DISC-06	PASS
	GEISA-API-DISC-07	PASS
	GEISA-API-DISC-08	PASS
GEISA API Instantaneous Requirements	GEISA-API-INST-01	PASS
GEISA API Sensor Requirements	GEISA-API-SENSOR-01	PASS
GEISA API Waveform Requirements	GEISA-API-WAVE-01	PASS
	GEISA-API-WAVE-02	PASS
	GEISA-API-WAVE-03	PASS
	GEISA-API-WAVE-04	PASS
	GEISA-API-WAVE-05	PASS

About this documentation

This documentation uses the AsciiDoc documentation generator. It is a convenient format that allows using plain-text formatted writing that can later be converted to various output formats such as HTML and PDF.

In order to generate an HTML version of this documentation, use the following command (the asciidoc package will need to be installed in your Linux distribution):

```
$ asciidoc test-report.adoc
```

This will result in a README.html file being generated in the current directory.

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```
$ asciidoctor-pdf test-report.adoc
```